

- 11 A satellite is orbiting the Earth with a radius of 6610 km at a speed of 7780 m s^{-1} . The satellite is boosted to a higher orbit of radius 6890 km. Given that the mass of the Earth is $6.0 \times 10^{24} \text{ kg}$, the speed of the satellite in the new orbit is

A 7460 m s^{-1} **B** 7620 m s^{-1} **C** 7940 m s^{-1} **D** 8110 m s^{-1}

Ans: B

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$\frac{GM}{r} = v^2$$

$$v^2 \propto \frac{1}{r}$$

$$\left(\frac{v_f}{v_i}\right)^2 = \frac{r_i}{r_f}$$

$$\left(\frac{v_f}{7780}\right)^2 = \frac{6610}{6890}$$

$$v_f = 7620 \text{ m s}^{-1}$$

Note that you cannot assume total energy of satellite remains the same as the satellite is 'boosted' to a higher orbit (meaning that there is an input energy).

- 12 A small ice cube of mass 20 g is heated and changes from the solid to the liquid state. During